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by [Mansour Saeed Mubarak Hamdan Al Hamdani](#) - Sunday, 10 August 2025, 5:45 PM

The Rise of Agent-Based Systems and Their Benefits to Organizations

The further onto the better path of development the aspect of agent-based systems has come owing to artificial intelligence (AI) developments, the increasing power of computation, and a need for decentralized decision-making. The rise in agent-based systems is attributed to the increasing complexity of modern organizations, which traditional centralized systems cannot adapt to. These systems are made up of autonomous agents that work together to solve problems, drawing from the AI perspective of multi-agent systems and machine learning (Wooldridge, 2009). The recent development of powerful simulation tools, alongside big data, has brought these systems into use across industries.

One of the advantages of using agent-based systems is decision making. In supply chain management, agents can handle real-time data and adapt to an ever-changing environment. Wooldridge and Jennings (1995) describe how agent-based systems perform logistics by letting agents negotiate and dynamically reroute deliveries. Such flexibility may minimize costs and maximize the agent's efficiency. Other advantages include the ability to permit fault tolerance so that if one agent fails, other agents can take over and keep the system functional (Jennings et al., 1998). This is especially useful in the case of critical infrastructures such as power grids, where reliability is of main concern.

Moreover, these ABS can foster innovation through collaborative problem-solving. In the health area, agent-based models simulate the spread of disease and help formulate policies. Macal and North (2010) studied the interventions of ABS in forecasting outbreaks of influenza and targeting interventions that saved resources. This particular strength highlights the transformative possibility of agent-based systems in reorienting the organizational landscape through scalable, data-driven insight.

In conclusion, the growth of agent-based systems was in light of technological advancement and organizational needs, boasting advantages of improved decision-making, fault tolerance, and innovation. Their footing in AI research will continue to cause transformations with more vigor, making them an absolute requirement in today's ever-changing climate. Theoretical and practical significance of these insights has been reaffirmed in the academic literature.

References

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Peer Response

by [Thiago Contardi](#) - Thursday, 14 August 2025, 3:55 PM

Your post gives a clear overview of how agent-based systems (ABS) have evolved and the benefits they bring to sectors like supply chain management and healthcare. While ABS can be robust in these scenarios, the accuracy and reliability of their outputs depend on the quality of the models and the data.



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